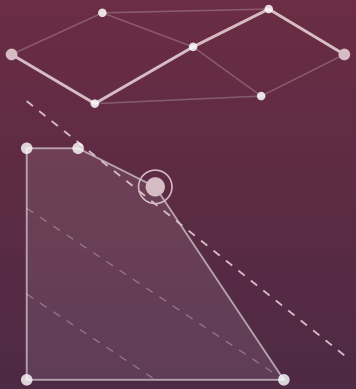
Operations Research

Lecture 1 · Semester 5



المدرسة الوطنية العليا لتكنولوجيا الأنظمة المستقلة

NHSAST

NATIONAL HIGHER SCHOOL OF AUTONOMOUS SYSTEMS TECHNOLOGY

Writing a Linear Program

From a problem in words to a mathematical model

NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

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Plan and goal

1. What is operations research?
2. A first example, step by step: the drone workshop
3. The words we use, and a method in five steps
4. Three more examples: mixing, transport, assignment
5. Different ways to write the same linear program
6. Useful tricks, and the limits of linear programs

Goal from a text, write a correct linear program.

What is operations research?

- Operations research (OR) uses mathematics to make the best decision when resources are limited (time, money, material, energy).
- Two steps: (1) write a model of the problem; (2) solve the model.
- Linear programming (LP): the model contains only linear expressions, like $40x_1 + 60x_2$.
- History: Kantorovich (1939), Dantzig and the simplex method (1947), Nobel Prize in Economics (1975).

Where is linear programming used?

- Production: how many units of each product should we make?
- Mixing: how much of each raw material should we put in a mixture?
- Transport: which warehouse sends goods to which customer?
- Planning over time: production, stock, stored energy
- Scheduling people and machines
- Robots and drones: which robot does which task, where to send spare parts, when to charge the batteries

A first example: the drone workshop

A small workshop assembles two models of drones: model A (simple) and model B (with a camera).

	Model A	Model B	Available per week
Assembly (hours for one drone)	1	2	40 hours
Testing (hours for one drone)	3	2	60 hours
Cameras (for one drone)	0	1	18 cameras
Profit (thousand DA for one drone)	40	60	

Question: how many drones of each model should be made each week to get the largest profit?

Question 1: what do we decide? $\rightarrow$ the variables

We must decide two numbers:

- x_1 = number of drones A made per week
- x_2 = number of drones B made per week

These unknowns are the decision variables. Drones are counted in whole numbers: we come back to this on slide 13.

A choice of values is a plan. Example: $(x_1, x_2) = (10, 5)$ means "10 drones A and 5 drones B per week".

Tip always write what a variable means, with its unit and its period.

Question 2: what do we want? $\rightarrow$ the objective

Profit $=$ 40 thousand DA for each drone A $+$ 60 thousand DA for each drone B:

$$z = 40x_1 + 60x_2 \qquad \text{(to maximise)}$$

- Example: for the plan $(10, 5)$, $z = 40 \cdot 10 + 60 \cdot 5 = 400 + 300 = 700$ thousand DA (700 000 DA).
- Units: (thousand DA per drone) $\times$ (drones) $=$ thousand DA.

z is the objective function. We want to maximise it (a profit) or to minimise it (a cost, a time, ...).

Question 3: what limits us? $\rightarrow$ the constraints

- **Assembly:** x_1 drones A use $1 \cdot x_1$ hours, x_2 drones B use $2 \cdot x_2$ hours. At most 40 hours:
$$x_1 + 2x_2 \leq 40$$
- **Testing:** $3x_1 + 2x_2 \leq 60$
- **Cameras:** one per drone B: $x_2 \leq 18$
- **Signs:** we cannot make a negative number of drones: $x_1 \geq 0$, $x_2 \geq 0$

These conditions are the constraints.

The complete model: a linear program

$$
\begin{aligned}
\max \quad & z = 40x_1 + 60x_2 && \text{(profit, thousand DA per week)} \\
\text{subject to} \quad & x_1 + 2x_2 \leq 40 && \text{(assembly hours)} \\
& 3x_1 + 2x_2 \leq 60 && \text{(testing hours)} \\
& x_2 \leq 18 && \text{(cameras)} \\
& x_1 \geq 0,\ x_2 \geq 0 && \text{(signs)}
\end{aligned}
$$

Careful: the *real problem* asks whole numbers, $x_1, x_2 \in \{0, 1, 2, \dots\}$. The LP above only asks $x_1, x_2 \geq 0$ (real numbers): it is the LP relaxation of the real problem (slide 13).

Definition 1.1 (linear program)

A linear program (LP) is: maximise or minimise a linear function of the variables, subject to linear constraints of the form "$\leq$", "$=$" or "$\geq$".

The words we use

Definition 1.2

- A feasible solution is a plan that satisfies *all* the constraints.
- The feasible set is the set of all feasible solutions.
- An optimal solution is a feasible solution with the best value of z; this best value is the optimal value.

- $(10, 10)$: $10 + 20 = 30 \leq 40$, $30 + 20 = 50 \leq 60$, $10 \leq 18$: feasible, $z = 1000$.
- $(15, 15)$: $15 + 30 = 45 > 40$: not feasible (one violated constraint is enough).

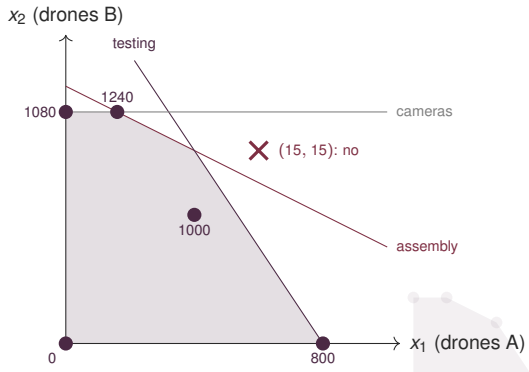
Testing some plans

Plan (x_1, x_2)	Assembly (≤ 40)	Testing (≤ 60)	Cameras (≤ 18)	Feasible?	Profit z
(0, 0)	0	0	0	yes	0
(20, 0)	20	60	0	yes	800
(0, 18)	36	36	18	yes	1080
(10, 10)	30	50	10	yes	1000
(15, 15)	45	75	15	no	—
(4, 18)	40	48	18	yes	1240

Profits in thousand DA. There are infinitely many feasible plans: we cannot test them all. We need a method (Lecture 2).

A plan is a point of the plane

- Each plan (x_1, x_2) is a point.
- Each constraint is a line; the feasible plans form the shaded region.
- The plans of the table: dots with their profit. $(15, 15)$ is outside: it violates assembly.
- Finding the best plan = finding the best point of the region: Lecture 2.



Whole numbers or real numbers?

- **Real problem:** $x_1, x_2 \in \{0, 1, 2, \dots\}$ (whole numbers of drones). This is an integer program.
- **LP relaxation:** $x_1, x_2 \geq 0$, any real values. Easier to solve (Lecture 2).
- Here we are lucky: the best LP plan is $(10, 15)$, with $z = 1300$ (Lecture 2). These are whole numbers, so it is also the best real plan.
- This luck is not guaranteed. With 61 testing hours instead of 60:

	Plan (x_1, x_2)	z (thousand DA)
Best LP plan	$(10.5,\ 14.75)$: not whole numbers	1305
Round up	$(11, 15)$: assembly $41 > 40$, not feasible	—
Round down	$(10, 14)$	1240
Best whole-number plan	$(10, 15)$	1300

What does “linear” mean?

Allowed (linear):

- a number times a variable: $40x_1$, $-2x_3$, $0.5x_2$
- sums of such terms: $40x_1 + 60x_2$
- a constant added: $3x_1 + 2x_2 + 10$ (strictly speaking affine)

Not allowed (not linear):

- products of variables: $x_1 x_2$
- powers: x_1^2, roots: $\sqrt{x_1}$
- division by a variable: $1/x_1$
- absolute values: $|x_1 - x_2|$

Comparison signs

An LP uses only $\leq, =, \geq$. $x_1 < 5$ is linear, but strict signs $<, >$ are not allowed: $\max\ x_1$ s.t. $x_1 < 5$ has no best plan (4.9, 4.99, 4.999, . . .).

Your turn · 1 min linear or not? $3x_1 - x_2 \leq 5$; $x_1 x_2 \geq 4$; $x_1 + 2 = 7x_2$.

Method 1.1: write a linear program in five steps

1. **Read** the text twice. Put all the numbers in a table, with units.
2. **Variables**: what do we decide? Write the meaning of each variable, with unit and period.
3. **Objective**: what do we want? Write z, and say $\max$ or $\min$.
4. **Constraints**: what limits us? One constraint for each limit, each requirement, each balance.
5. **Check**: sign constraints, units, every number of the text used, everything linear.

From words to constraints

In the text	Constraint
“at most b”, “not more than b”, “we have b”	$\cdots \leq b$
“at least b”, “not less than b”, “we need b”	$\cdots \geq b$
“exactly b”, “all of it must be used”	$\cdots = b$
“x is at least twice y”	$x \geq 2y$, i.e. $x - 2y \geq 0$
“at least 30% of the total is x_1”	$x_1 \geq 0.3\,(x_1 + x_2 + x_3)$ (Rule 1.1)
“x and y differ by at most 5”	$x - y \leq 5$ and $y - x \leq 5$ (Rule 1.7)
“what comes in = what goes out”	a balance: $\cdots = \ldots$

Example 1.2: mixing — the problem

A factory makes exactly 1000 kg of a special material for robot arms, by mixing three raw materials M1, M2 and M3. Each raw material contains two important ingredients, A and B. The final material must contain between 18% and 20% of ingredient A, and at least 8% of ingredient B.

Raw material	Ingredient A	Ingredient B	Cost (DA/kg)	Available
M1	10%	2%	80	no limit
M2	5%	30%	600	300 kg
M3	60%	0%	250	no limit

Find the cheapest mixture. **Your turn · 2 min** variables and objective.

Example 1.2: mixing — the model

x_1, x_2, x_3 = kilograms of M1, M2 and M3 in the mixture.

- Ingredient A in the mixture: 10% of x_1 + 5% of x_2 + 60% of x_3
  $= 0.10x_1 + 0.05x_2 + 0.60x_3$ kg.
- 18% of 1000 kg $= 180$ kg; 20% of 1000 kg $= 200$ kg; 8% of 1000 kg $= 80$ kg.

$$
\begin{aligned}
\min \quad & z = 80x_1 + 600x_2 + 250x_3 && \text{(cost, DA)} \\
\text{s.t.} \quad & x_1 + x_2 + x_3 = 1000 && \text{(total mass)} \\
& 180 \le 0.10x_1 + 0.05x_2 + 0.60x_3 \le 200 && \text{(ingredient A: two constraints)} \\
& 0.02x_1 + 0.30x_2 \ge 80 && \text{(ingredient B)} \\
& x_2 \le 300, \quad x_1, x_2, x_3 \ge 0
\end{aligned}
$$

Rule 1.1: percentages of a total

If the total mass is not fixed, “at least 18% of ingredient A” first gives a quotient:

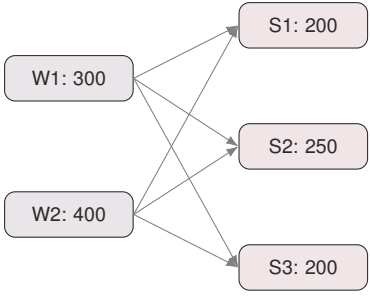
$$\frac{0.10x_1 + 0.05x_2 + 0.60x_3}{x_1 + x_2 + x_3} \geq 0.18 \qquad \text{(not linear)}$$

Rule 1.1

The total is positive, so we can multiply both sides by the total (the inequality keeps its direction), then bring everything to the left.

$$\begin{aligned} & 0.10x_1 + 0.05x_2 + 0.60x_3 \geq 0.18x_1 + 0.18x_2 + 0.18x_3 \\ \iff \quad & -0.08x_1 - 0.13x_2 + 0.42x_3 \geq 0 \qquad \text{(linear)} \end{aligned}$$

Example 1.3: transport — the problem



Two warehouses W1, W2 store spare batteries; three robot sites S1, S2, S3 need them.

Cost (DA/battery)	S1	S2	S3	Stock
W1	800	600	1000	300
W2	900	1200	700	400
Need	200	250	200	

Find the cheapest transport plan. What do we decide?

Example 1.3: transport — the model

x_{ij} = number of batteries sent from warehouse Wi to site Sj $(i = 1, 2;\ j = 1, 2, 3)$: six variables.

$$\begin{aligned}
\min \quad & z = 800x_{11} + 600x_{12} + 1000x_{13} + 900x_{21} + 1200x_{22} + 700x_{23} \\
\text{s.t.} \quad & x_{11} + x_{12} + x_{13} \le 300, \qquad x_{21} + x_{22} + x_{23} \le 400 && \text{(stocks)} \\
& x_{11} + x_{21} \ge 200, \quad x_{12} + x_{22} \ge 250, \quad x_{13} + x_{23} \ge 200 && \text{(needs)} \\
& x_{ij} \ge 0
\end{aligned}$$

Short form: $\min \sum_i \sum_j c_{ij} x_{ij}$ s.t. $\sum_j x_{ij} \le s_i$, $\sum_i x_{ij} \ge d_j$, $x_{ij} \ge 0$ ($\sum$ means “sum”).

Example 1.3: a simple check before solving

- Total stock: $300 + 400 = 700$ batteries. Total need: $200 + 250 + 200 = 650$.
- $650 \leq 700$: there is enough stock, so a feasible plan exists.
- Example of feasible plan: W1 sends 200 to S1 and 100 to S2; W2 sends 150 to S2 and 200 to S3. Cost $= 160\,000 + 60\,000 + 180\,000 + 140\,000 = 540\,000$ DA.
- If the sites needed 800 batteries in total, no plan could satisfy all the needs: the problem would be infeasible.

Example 1.4: assignment — the problem

Three robots R1, R2, R3 must do three tasks T1, T2, T3: each robot does one task, and each task is done by one robot.

Time (minutes)	T1	T2	T3
R1	9	2	7
R2	6	4	3
R3	5	8	1

Find the assignment with the smallest total time.

Your turn · 1 min how many possible assignments are there?

Example 1.4: assignment — the model

$x_{ij} = 1$ if robot Ri does task Tj, and $x_{ij} = 0$ otherwise (nine variables).

$$\min\ z = 9x_{11} + 2x_{12} + 7x_{13} + 6x_{21} + 4x_{22} + 3x_{23} + 5x_{31} + 8x_{32} + 1x_{33}$$

- Each robot does one task: $x_{11} + x_{12} + x_{13} = 1,\ x_{21} + x_{22} + x_{23} = 1,\ x_{31} + x_{32} + x_{33} = 1$
- Each task is done by one robot: $x_{11} + x_{21} + x_{31} = 1,\ x_{12} + x_{22} + x_{32} = 1,$ $x_{13} + x_{23} + x_{33} = 1$
- $x_{ij} \geq 0$

- “0 or 1” is not a linear condition: we write only $x_{ij} \geq 0$.
- We accept this fact: for this problem, the LP always has an optimal solution with values 0 and 1.

Writing an LP with matrices

Example 1.1 (drone workshop):

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad c = \begin{pmatrix} 40 \\ 60 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \\ 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 40 \\ 60 \\ 18 \end{pmatrix}$$

The model is: $\max\ c^\top x \quad \text{s.t.} \quad Ax \le b,\ x \ge 0.$

- $c^\top x = 40x_1 + 60x_2; \quad Ax = \begin{pmatrix} x_1 + 2x_2 \\ 3x_1 + 2x_2 \\ x_2 \end{pmatrix}.$
- $Ax \le b$ is read line by line: one line of A = one constraint.
- One column of A = what one unit of a variable uses (column 1: drone A uses 1 h, 3 h, 0 camera).

Two standard ways to write an LP

Definition 1.3 (canonical form)

$\max\ c^\top x \quad \text{s.t.} \quad Ax \le b,\ x \ge 0$ (all constraints "$\le$", all variables ≥ 0)

Definition 1.4 (standard form)

$\max\ c^\top x \quad \text{s.t.} \quad Ax = b,\ x \ge 0$ (all constraints "$=$", all variables ≥ 0)

- The drone workshop is already in canonical form.
- The standard form is the form used by the simplex method (later in the course).
- Every LP can be written in both forms, with the rules of the next slide.

Rules 1.2 to 1.6: changing the form

Rule	If we have	we write	Example
1.2	min z	max $(-z)$	min $3x_1 + x_2 \to$ max $-3x_1 - x_2$
1.3	a constraint "$\geq$"	multiply by -1: "$\leq$"	$x_1 + x_2 \geq 2 \to -x_1 - x_2 \leq -2$
1.4	a constraint "$=$"	two constraints "$\leq$" and "$\geq$"	$x_1 - x_2 = 1 \to x_1 - x_2 \leq 1,\ x_1 - x_2 \geq 1$
1.5	a free variable x_j	$x_j = x_j^+ - x_j^-,\ x_j^{\pm} \geq 0$	$x_2 = 3$: $x_2^+ = 3,\ x_2^- = 0$
1.6	a constraint "$\leq$"	add a slack $s \geq 0$	$x_1 + 2x_2 \leq 40 \to x_1 + 2x_2 + s_1 = 40$
	a constraint "$\geq$"	subtract a surplus $e \geq 0$	$x_1 + x_2 \geq 2 \to x_1 + x_2 - e_1 = 2$

Rule 1.2: the best plan is the same; only the sign of the optimal value changes.

Two worked examples

(a) Drone workshop in standard form (slacks s_1, s_2, s_3 = unused hours and cameras):

$$\max\ 40x_1 + 60x_2$$

$$\text{s.t. } x_1 + 2x_2 + s_1 = 40, \quad 3x_1 + 2x_2 + s_2 = 60, \quad x_2 + s_3 = 18, \quad x_1, x_2, s_1, s_2, s_3 \geq 0$$

For the plan $(10, 10)$: $s_1 = 40 - 30 = 10$ unused assembly hours, $s_2 = 10$, $s_3 = 8$.

(b) $\min\ 3x_1 + 2x_2$ s.t. $x_1 + x_2 \geq 2$, $x_1 - 2x_2 \leq 4$, $x_1 \geq 0$, x_2 free.

Canonical form: $\max\ -3x_1 - 2x_2^+ + 2x_2^-$

$$\text{s.t. } -x_1 - x_2^+ + x_2^- \leq -2, \quad x_1 - 2x_2^+ + 2x_2^- \leq 4, \quad x_1, x_2^+, x_2^- \geq 0$$

Best plan: $(0, 2)$, $z = 4$. Warning: changing the form does not tell if a best plan exists. With $\min\ 3x_1 - x_2$, the same constraints give an unbounded LP: $x_1 = 0$, $x_2 = t \geq 2$ gives $z = -t \to -\infty$.

Rules 1.7 and 1.8: two useful tricks

Rule 1.7 (absolute value $\leq$)

$|x_1 - x_2| \leq 5 \iff x_1 - x_2 \leq 5$ and $x_2 - x_1 \leq 5$. Check: $x = (8, 2)$: $|6| = 6 > 5$, and $8 - 2 = 6 > 5$.

Rule 1.8 (maximise the smallest value)

$\max \min(x_1, x_2) \longrightarrow$ add a variable t: $\max\ t$ s.t. $t \leq x_1,\ t \leq x_2$ (and the other constraints).

Not possible: a product like $x_1 x_2 \leq 1$. The points $(2, 0.5)$ and $(0.5, 2)$ satisfy it, but their middle point $(1.25, 1.25)$ does not ($1.5625 > 1$); a set defined by linear constraints always contains the middle point of two of its points.

Method 1.2: what to do with a strict inequality

Whole numbers (we count objects):
the rewriting is exact.

- fewer than 25 drones:
  $x_1 + x_2 < 25 \;\rightarrow\; x_1 + x_2 \leq 24$
- more drones A than B:
  $x_1 > x_2 \;\rightarrow\; x_1 - x_2 \geq 1$
- some drones B: $x_2 > 0 \;\rightarrow\; x_2 \geq 1$

Real numbers (hours, kg, kWh):

1. Replace $<$ by $\leq$, and solve.
2. The best plan satisfies $<$: done, it is also the best plan.
3. The best plan gives $=$: no best plan. If b is really forbidden, use a margin $\varepsilon > 0$: $\leq b - \varepsilon$.

Example: assembly strictly less than 40 hours

Best plan $(10, 15)$ uses exactly 40 h: no best plan. $(9.9,\ 14.85)$ gives 1287, $(9.99,\ 14.985)$ gives 1298.7, ... never 1300. With $\varepsilon = 0.5$ h: $x_1 + 2x_2 \leq 39.5$, best plan $(10.25,\ 14.625)$, $z = 1287.5$.

The four hypotheses of linear programming

- **Proportionality**: 2 drones bring twice the profit of 1 drone (no discount for large quantities).
- **Additivity**: the total is the sum of the parts (the profit of A does not depend on B).
- **Divisibility**: the variables can take any real value (10.5 drones is allowed: the LP relaxation, slide 13).
- **Certainty**: all the numbers of the model are known exactly.

If we need whole numbers: integer programming. If the numbers are uncertain: sensitivity analysis (Chapter 3).

Summary and next lecture

- An LP: variables, a linear objective ($\max$ or $\min$), linear constraints, sign constraints.
- Method 1.1: read, variables, objective, constraints, check.
- Examples: production, mixing, transport, assignment. Whole numbers: the LP is a relaxation.
- Matrix form; canonical form ($\leq$) and standard form ($=$); Rules 1.2 to 1.6 to change the form.
- Tricks: percentages (Rule 1.1), absolute values (Rule 1.7), max of the smallest (Rule 1.8). Strict inequalities: Method 1.2.
- TD 1: modelling exercises. Next lecture: how to solve an LP, and all the possible cases.